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BIOTECHNOLOGY and BIOENGINEERING

# A Label-Free Methodology for Selective Protein Quantification by Means of Absorption Measurements

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Received 1 March 2011; revision received 18 May 2011; accepted 25 May 2011

Published online 13 July 2011 in Wiley Online Library (wileyonlinelibrary.com). DOI 10.1002/bit.23229

**ABSTRACT:** The application of high throughput experimentation (HTE) in protein purification process development has created an analytical bottleneck. Using a new label-free and non-invasive methodology for analyzing multicomponent protein mixtures by means of spectral measurements, we show that the analytical throughput for selective protein quantification can be increased significantly. An analytical assay based on this new methodology was shown to generate very precise results. Further, the assay was successfully applied as analytics for a resin screening performed in HTE mode. The increase in analytical throughput was obtained without decreasing the level of information when compared to analytical chromatography. This proves its potential as a valuable analytical tool in conjugation with high throughput process development (HTPD). Further, fast selective protein quantification can enhance process control in a commercial production environment and, hence, minimize the need for off-line release analysis.

Biotechnol. Bioeng. 2011;108: 2661–2669.



**KEYWORDS:** protein analytics; specific protein quantification; label-free; non-invasive; partial least squares; high throughput process development

## Introduction

Over the past few years, the use of high throughput techniques for development of protein purification processes has increased strongly. The main drivers of this shift toward high throughput process development (HTPD) have been (and still are) the need to speed up both time to clinic and time to market. Automated performance of unit operations in a miniaturized and parallel mode has delivered the high experimental throughput which is the basis of HTPD. The automation has been realized using robotic workstations with liquid handling systems where sampling is commonly performed in microtiter plates.

Alongside the development of high throughput experimentation (HTE) techniques, a trend of replacing the traditional "one factor at the time" screenings with more sophisticated approaches such as factorial designs, intelligent search algorithms, and mechanistic modeling has evolved (Degerman et al., 2009; Oelmeier et al., 2011; Susanto et al., 2009). These approaches make process development more efficient and can result in a deeper understanding of the developed processes. However, regardless of how sophisticated a search algorithm or design is used to improve the experiment to information ratio, a high number of samples have to be analyzed when performing state of the art HTPD. In order not to compromise the throughput and level of automation, univariate spectroscopic measurements performed in plate readers is the most comfortable analytical solution. If the nature of the experiments is more complex and requires selective or specific protein quantification, analytical chromatography, mass spectrometry, electrophoresis, and/or enzyme linked immunosorption assay can be applied. However, all methods have drawbacks related to throughput, sample preparation, and/or automated integration. Hence, the enhanced experimental throughput often creates an analytical bottleneck where the experimenter is forced to make decisions on the basis of a trade off between analysis time and analytical information.

Liquid chromatography is the main workhorse of modern protein purification. Accordingly, liquid chromatography, in batch as well as column mode, is among the unit operations readily established and applied in HTPD (Coffman et al., 2008; Wierling et al., 2007; Wiendahl et al., 2008). Batch experiments can give information on selectivity, uptake kinetics, and static binding capacities in

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studies of different phase combinations. From column experiments, information on dynamic binding capacities and chromatographic resolution can be derived. Both batch and column chromatography in HTE mode can also be used to determine parameters for mechanistic modeling of chromatographic processes (Susanto et al., 2008).

Column chromatography in HTE mode generates hundreds of samples per hour and hence represents a major analytical challenge if the high experimental throughput is to be maintained. The objective of the work presented here, was to develop and establish an assay which would limit the analytical trade off without compromising the throughput of the experimental performance. Such an assay would in general be useful to unit operations relevant to protein purification where selective protein quantification is necessary: e.g., membrane chromatography, batch chromatography, selective protein precipitation, protein crystallization, and aqueous two-phase systems.

The standard univariate spectroscopic assay used to quantify proteins in aqueous solutions is based on the application of the Lambert–Beer law to a measured UV absorption caused by solute protein molecules. In most cases, the absorption ascribable to the aromatic structures in the residues tryptophan, tyrosine, and phenylalanine is used. Of these amino acids, tryptophan and tyrosine have far the strongest absorption. Figure 1 displays the absorption spectra in the range from 220 to 300 nm of pure tryptophan, tyrosine, and phenylalanine in aqueous solutions. Their spectra are characterized by two maxima in the mid-UV range: the most intense around 230 nm and one of lower intensity and more wide stretching around 280 nm. The absorptivity of phenylalanine is approximately one magnitude lower and both maxima are shifted 20 nm toward lower wavelengths. Due to material and apparatus issues, it is often preferred to measure the absorption at 280 nm. However, as

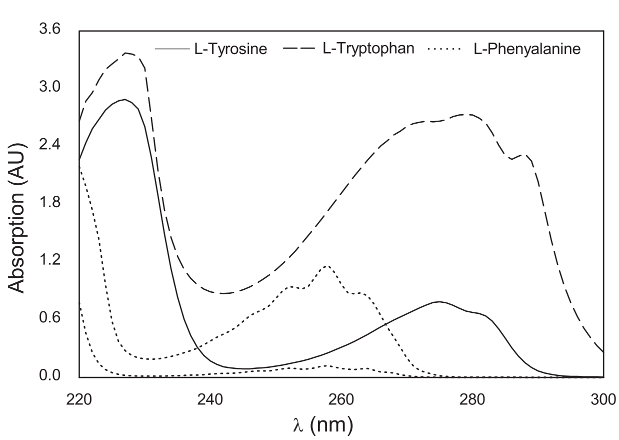


**Figure 1.** Mid-UV absorption spectra of tyrosine, tryptophan, and phenylalanine. The concentration of tyrosine and tryptophan was 0.1 g/L. Due to the relative low absorptivity of phenylalanine, absorption spectra of both the concentrations 1.0 g/L and 0.1 g/L are displayed.

the extinction coefficient (ε) differs depending on the amount and kind of aromatic residues present in a protein, the method is only applicable to pure protein solutions. For absolute quantification of proteins with unknown extinction coefficients or protein mixtures, methods combining the absorption at 280 with the absorption at 205 nm where also the peptide bonds absorb can be found in the literature (Scopes, 1974; Whitaker and Granum, 1980).

Mid-UV spectral data have been applied for identification and selective quantification of many different compounds other than proteins: for example (Muñoz-Muñoz et al., 2010; Thoma and Ziegler, 1998; Xie et al., 1999). However, until now mid-UV absorption spectra of proteins have mainly been applied to monitor structural and binding aspects of proteins: for example (Anantharamaiah et al., 1999; Krzyzanowska et al., 1998; Rock, 1983). One exception is the application of UV absorption spectra to determine protein and nucleic acid content of viruses (Porterfield and Zlotnick, 2010). As can be seen in Figure 1, the absorption spectra of the aromatic amino acids differs significantly, not only in intensity but also in shape. This fact will cause the absorption spectra of different proteins to change depending on the ratios of the aromatic residues present in the proteins. Awareness of this fact combined with the spreading application of chemometric approaches (e.g., multivariate calibration) in biomedical analysis (Escandar et al., 2006) has been the inspiration to a new methodology for analyzing multi-component protein mixtures in a label-free and non-invasive mode. According to the definition by Lavine (2000), chemometrics is an approach to analytical and measurement science that uses mathematical, statistical, and other methods of logical science to determine properties of substances that would otherwise be difficult to determine directly. In this case the substance property is the selective protein concentration in a mixture of several proteins and the applied "logical science" is *partial least squares—projection to latent structures* (PLS) regression. This was chosen as PLS is known to be particular good in coping with expected collinearity of the spectral data (Næs and Mevik, 2001).

## Materials and Methods

### Disposables

*Sample preparation*: 96-well square deep well plates (DWP), volumetric capacity 2 mL, polypropylene (VWR, Germany).

*Mid-UV spectra measurement*: 96-well half area UV microtiter plates (MTP), volumetric capacity 180 μL, polystyrene, flat bottom, and 384 square well MTP, volumetric capacity 110 μL, polystyrene, flat bottom (Greiner, Germany).

*Chromatography*: RoboColumns purchased prepacked with 200 μL of the following resins: CM Ceramic HyperD F, Toyopearl SP 650 M, SP Sepharose FF, and Fractogel EMD $SO_3$ (Atoll, Germany).

## Model Proteins

The following model proteins were used: lysozyme (lys) from hen egg white, cytochrome C (cytC) from horse heart, and ribonuclease A (ribA) from bovine pancreas (Sigma–Aldrich, St. Louis, MO). These proteins compose a convenient model system for method development of protein purification due to the distribution of their isoelectric points. At neutral pH, separation can be achieved using cation exchange chromatography.

## Liquid Handling Station

The generation of protein samples for model calibration, model validation, and elution buffers as well as the chromatographic experiments were performed automated on a Tecan Freedom Evo 200 (Tecan, Crailsheim, Germany) liquid handling station (LHS). To enable performance of column chromatography on the LHS, a custom made column carrier was used. A similar carrier is available under the name Te-Chrom (Tecan). A thorough description of parallelized chromatography performed automated on a LHS has been published by Wiendahl et al. (2008) and Susanto et al. (2008). Further, the LHS was equipped with an Infinite 200 plate reader (Tecan) to measure the mid-UV spectra of the calibration and validation samples as well as the chromatographic fractions. The LHS was operated with EVOware 2.0. Import of values for pipetting volumes and export of spectral data was handled via Excel (Microsoft, Redmond, WA).

## Experimental Setup

### Model Calibration and Model Validation

The calibration and validation samples were based on a four level D-optimal onion design generated with MODDE (Umetrics, Umeå, Sweden). In total, the four level onion design with three center points consisted of 29 samples. The design covered 19 mixing ratios and nine concentration levels from 0 to 0.7 g/L. In Figure 2A the mixing ratios of all three proteins and the absolute concentrations of ribA and cytC are displayed. An additional concentration level at 0.05 g/L was added to the four layer onion design, however only as pure samples. In total a sample set consisted of 32 samples. Two milliliters of each sample were prepared from stock solutions, containing 2.1 g/L protein in a sodium phosphate buffer at pH 7. Preparing 2 mL of each sample, enabled testing of different plate types and sample volumes based on the same sample stock. For the measurement of the absorption spectra, a defined volume of the prepared samples were transferred to UV transparent MTPs. The absorption spectra were measured with 1 nm resolution in the range 240–300 nm. For model calibration based on the obtained spectral data, the MATLAB based PLS toolbox (Eigenvector Research, Wenatchee, WA) was used. The spectral data was preprocessed by mean centering which is done by subtracting the mean spectrum of all calibration samples from each protein spectrum. Method validation was performed according to the flow scheme displayed in Figure 2. The setup was designed to obtain a reliable measure of the possible obtainable precision taking into account pipetting errors, measurement error, and errors caused by plate deviations. The setup consisted of six independently pipetted sample sets (Fig. 2C and E). Five sample sets were used to calibrate five models (Fig. 2D). All five calibrated models were then used to experimentally determine the concentrations of the sixth sample set (Fig. 2E). From these values, absolute errors, relative errors, and 95% confidence intervals were calculated (Fig. 2F–H).

### Resin Screening

All columns were equilibrated with 5 CV (1 CV = 200 μL) loading buffer (20 mM sodium phosphate, pH = 7) before loading them with 0.25 mg of each protein per column. This protein load equals 3.75 g/L resin of total protein. Before starting the actual column experiments, 24 different buffer solutions with increasing salt concentrations were prepared and stored in DWPs until use. The prepared buffer solutions composed a linear step gradient from 0 to 500 mM NaCl and was used to elute the proteins in 24 CV. Each gradient step was collected as one fraction. After completed elution, 100 μL of each elution fraction were transferred into UV grade MTPs and the absorption spectra were measured. Based on these spectra, the protein content in the samples was determined.

### Peak Fitting and Calculation of Chromatographic Resolution

To evaluate the chromatographic separations, an asymmetric Gaussian function was fitted to the data points. The fitting was performed using the graphing and data analyzing software OriginPro8.5 (OriginLab Corporation, Northampton, MA) which has an asymmetric Gaussian fit function under the name *bigauss*. The *bigauss* fitfunction is a two-step function which separately describes the leading and the trailing edges of an asymmetric peak using two different gauss curves. The bigauss fitfunction returns the values $w_1$ (FWHM/2—left side fit), $w_2$ (FWHM/2—right side fit), $x_c$ (retention at peak maximum), $H$ (maximal peak height), and $y_0$ (base line) for each peak. Based on the parameters $w_1$, $w_2$, and $x_c$ the resolution of two adjacent peaks $p_1$ and $p_2$ was calculated according to:

$$R_{p_1,p_2} = \frac{2.345(x_{c,p_2} - xc_{c,p_1})}{4(w_{2,p_1} + w_{1,p_2})} \qquad (1)$$

The above described equation is a derivative of the standard equation for chromatographic resolution based on

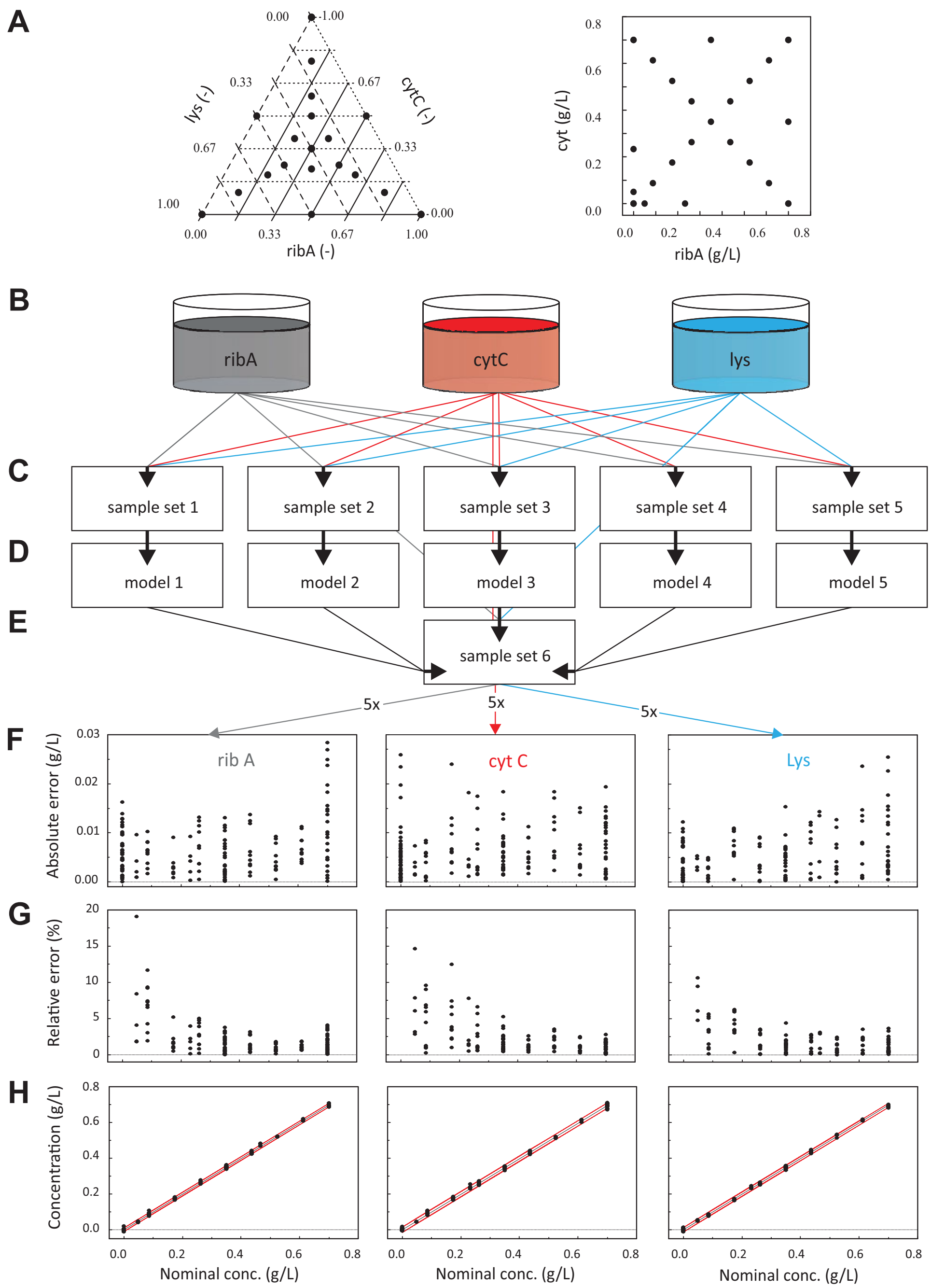


**Figure 2.** Design of the model calibration and validation samples followed by a flowchart describing the validation procedure. **A**: Mixing ratios of all three proteins and absolute concentrations of cytC and ribA. **B**: Protein stock solutions **C**: Five identical sample sets prepared individually from the protein stock solutions. **D**: Five calibrated models based on the five sample sets. **E**: Sixth sample set of which the concentrations were experimentally determined using the five calibrated models. **F**: Absolute errors of the experimentally determined concentrations displayed as a function of the nominal concentration. **G**: Relative error of each experimentally determined protein concentration as a function of the nominal concentration. **H**: Experimentally determined protein concentration plotted as a function of the nominal concentration. The red lines display 95% confidence interval for each protein species calculated on basis of the experimentally determined values. [Color figure can be seen in the online version of this article, available at http://wileyonlinelibrary.com/bit]

peak basis width at $4\sigma$ ($w_b$) and the distance between peak maxima ($t_{R_2}-t_{R_1}$):

$$R_s = \frac{2(t_{R_2}-t_{R_1})}{w_{b_1}+w_{b_2}} \quad (2)$$

### *Analytical Chromatography*

For comparison, the protein concentrations in the samples generated in the resin screening were quantified using a Mono S 4.6/100 PE column (GE Healthcare, Little Chalfont, Buckinghamshire, United Kingdom). The column was loaded with 20 μL sample and eluted with a multi linear sodium chloride gradient. Buffer A: 20 mM sodium phosphate, pH 7, buffer B: 20 mM sodium phosphate, 500 mM sodium chloride, pH 7. Gradient: buffer B 0–20% from 0 to 8 min, buffer B 20% from 8 to 12 min, buffer B 20–40% from 12 to 18 min, buffer B 100% from 18 to 20 min, buffer B 0% from 20 to 25 min. The flow rate was 1 mL/min.

## Results and Discussion

In the presented work, we describe a fast and automated assay for selective protein quantification. The label-free and non-invasive methodology on which the assay is based, uses PLS to correlate absorption spectra to selective protein concentrations. The developed assay has been subjected to a thorough validation and the results hereof will be presented and discussed. Further, an application relevant to HTPD is presented and the results are compared to results generated using standard analytical chromatography.

### Protein Mid-UV Absorption Spectra

In general, proteins absorb UV light in the mid-UV range. This absorption is mainly caused by the chromophoric properties of the three aromatic residues; tryptophan, tyrosine, and phenylalanine. Further, some proteins exhibit additional absorption in the visible range due to other chromophoric properties; for example, green fluorescence protein ($\lambda_{max} \sim 488$ nm) and cytochrome C ($\lambda_{max} \sim 526$ nm). Present in a mixture, these proteins can easily be quantified separately by measuring the absorption at wave lengths specific for each protein. In contrast, specific quantification of proteins with absorption characteristics only ascribable to their aromatic residues and peptide bonds, cannot be performed by such a straight forward method. However, the ratio and number of aromatic residues will vary among different proteins, and therefore the intensity and shape of the related UV absorption spectrum will vary.

Figure 3A displays the absorption spectra of the three model proteins ribA, cytC, and lys in the range 240–300 nm at a concentration of 0.5 g/L. To depicture the extend to which the spectral shapes vary independently of spectral intensity, difference spectra based on normalized spectra were calculated (Fig. 3B). The normalization was based on the maximal absorption value in the range 270–290 nm which was set to 1.0 AU. Figure 3C displays the difference spectra build by subtracting the mean of all three normalized spectra from each individual normalized spectrum. Lys and ribA have similar spectral shapes in the range 250–280 nm and deviates in the ranges 240–250 and 280–300 nm. The spectrum of cytC is different from both other spectra with the exception of the lys spectrum in the range 280–290 nm.

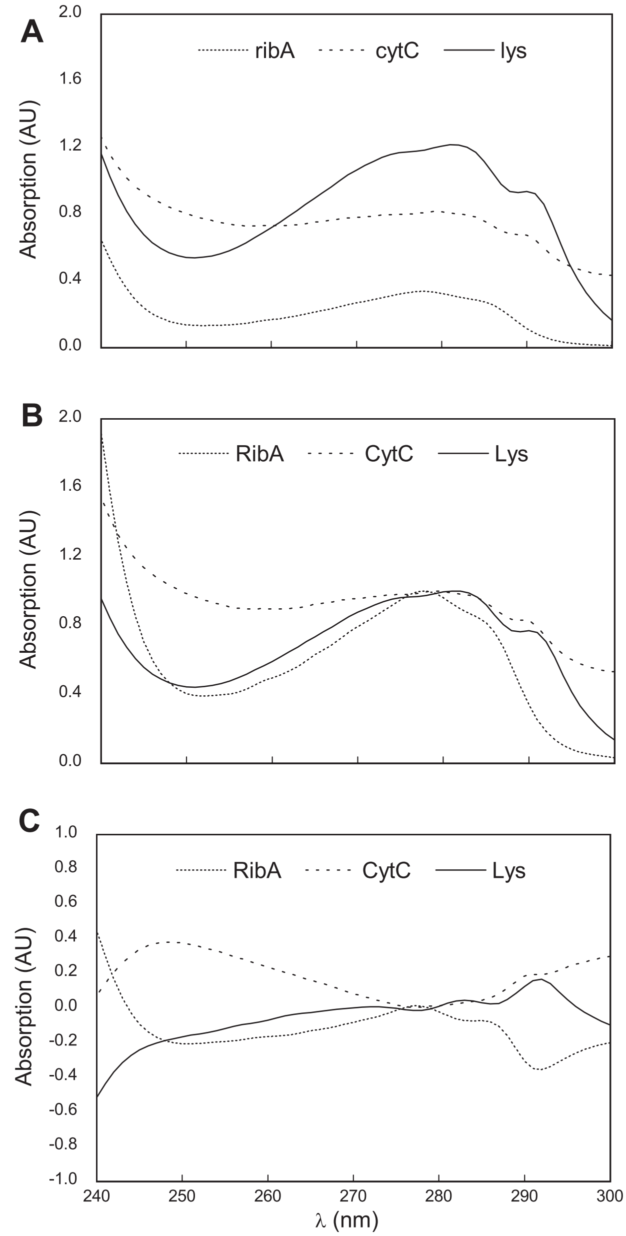


**Figure 3.** Absorption spectra of pure protein solutions measured at a concentration of 0.5 g/L and a light path length of 1 cm. **A**: original spectra **B**: normalized spectra **C**: difference spectra.

### Method Validation

A multivariate calibration model links a relation between several input variables to one or more properties of interest. In this case the input variables are the spectral data and the properties of interest are the selective concentrations of the different proteins contained in a sample. To determine the possible precision which can be obtained for the correlation of spectral data and protein concentrations using PLS, a thorough validation was undertaken according to the scheme presented in Figure 2. As a measure for precision, the 95% confidence interval was used. The confidence interval was based on the fivefold experimental determination of the protein concentrations in the samples of *sample set 6* (see Fig. 2).

In the following the validation results are presented. This includes the influence of plate type and sample volume, which in MTPs determines the optical path length, method robustness regarding salt concentration and temperature, in silico parameters such as choice of preprocessing and number of latent variables used for calibration, and finally PLS was compared to multiple linear regression (MLR).

### *Number of Latent Variables*

PLS calibration is based on a data reduction with the objective to capture only the significant variation needed for calibration. Initially, it was tested how the level of reduction would influence precision of the resulting PLS model. Here, the level of reduction is expressed by the number of latent variables included in the calibration of the PLS model where a low number of latent variables corresponds to a high data reduction. For each tested number of latent variables and for each protein species, the 95% confidence interval was calculated. Further, the confidence intervals for all three proteins were summarized for each number of latent variables to give a measure for the overall precision of the model. In Figure 4A the confidence intervals are displayed as a function of the number of latent variables in the tested range from 3 to 11. The curves fitted to the data points serves the visualization. Whereas cytC describes a low but continuous increase over the tested range, both lys and ribA exhibit a minimum at six latent variables. This is consistent with the fact that cytC differs the most from the mean spectrum of all three proteins (see Fig. 3). Hence, the most significant variation in the data is correlated to the cytC concentration and this variation will then be described more accurately by the first latent variables compared to ribA and lys. The sum of the confidence intervals is lowest for six latent variables. However, using six latent variables is a compromise, as the precision of cytC is not optimal at this point. Alternatively, a separately calibrated model for cytC could be used. Due to the very small difference in precision, it was chosen to base further method validation on models calibrated for all proteins based on six latent variables.

### *Plate Type and Sample Volume*

The influence of plate type and sample volume was tested by comparing the resulting confidence intervals. Included were two MTP types: 384 well MTP (square wells) and 96 well MTP (half area wells). In the 384 well MTP the tested volumes were 40, 60, and 80 μL corresponding to path lengths of 3.3, 5.5, and 6.7 mm. In the 96 well MTP the tested volumes were 40, 75, 100, and 150 μL corresponding to path lengths of 2.6, 4.7, 6.2, and 9.0 mm. The influence of sample volume and plate type is displayed in Figure 4B. To simplify the visualization, only mean values and standard deviations of the confidence intervals of all three proteins are displayed. In the range 2.6–5.0 mm, the optical path length correlated with a linear decrease of the confidence interval, independently of plate type. Above 5.0 mm the confidence interval decreased further when using the 96 well MTP, however the variance (expressed by the standard deviation) of the confidence intervals increased for the highest measured path length of 9.0 mm. Hence, it was chosen to perform all other

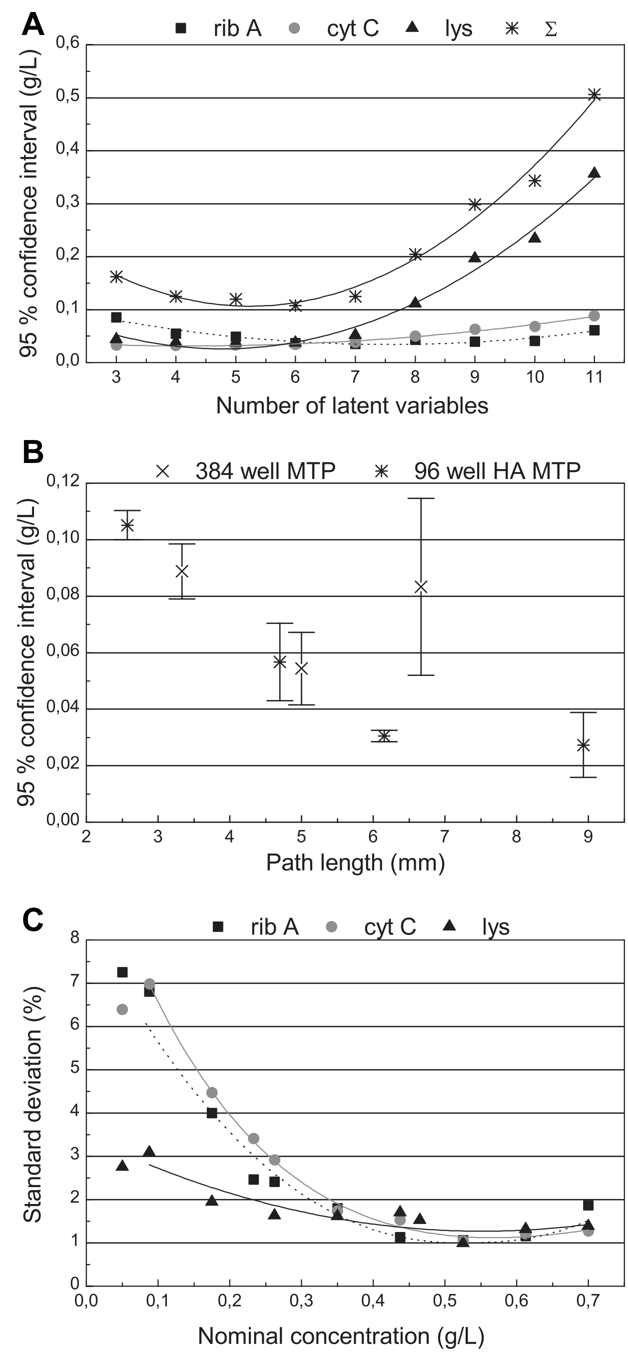


**Figure 4.** Results of the method validation. The curves fitted to the data points only serve the visualization of the data point trends. **A**: 95% confidence intervals of the experimentally determined protein concentrations (for each protein species and their sum) plotted as a function of the number of latent variables included in the PLS model. **B**: Mean confidence intervals and standard deviations as a function of the optical path length through the sample. **C**: Relative standard deviations of the experimentally determined protein concentrations as a function of the nominal concentration.

experiments with 100 μL sample in a 96 well MTP where the variance of the single confidence intervals was low (ribA: 0.031 g/L, cytC: 0.032 g/L, lys: 0.028 g/L) and the mean value was not significantly higher compared to what was achieved with 150 μL (0.031 vs. 0.027 g/L).

### Protein Concentration Range

The precision over the calibrated concentration range is displayed in Figure 4C (100 μL samples measured in 96 well MPTs and PLS models calibrated with 6 latent variables). The curves fitted to the data points serve to visualize the observed trends. They do not include the lowest concentration of 0.05 g/L, as this concentration level was only included in the form of pure samples and therefore not representative for complex samples. The precision increased with increasing protein concentrations up to 0.525 g/L, which is expressed by the decreasing relative standard deviation. For ribA and cytC the relative standard deviation decreased from 7% at 0.05 g/L to 1% at 0.525 g/L. Over the same range, the relative standard deviation for lys decreased from 3% to 1%. Above a concentration of 0.525 g/L, the relative standard deviation for all three protein species increased. The descending order of precision within the protein species (lys > ribA > cytC) correlated with the previously determined confidence intervals based on the same data (lys: 0.028 g/L, ribA: 0.031 g/L, cytC: 0.032 g/L). The initial decrease of the relative standard deviation is explained by pipetting errors and measurement noise having the highest impact at low concentrations. The increase of the relative standard deviation above 0.525 g/L is also expectable, since only a very low amount of light passes the sample at high protein concentrations. The lower limits of quantification (determined as $6\sigma$) were 0.04, 0.06, and 0.03 g/L for ribA, cytC, and lys, respectively. This is not immediately consistent with the relative standard deviations at the lowest concentration level presented in Figure 4C. However, as already mentioned, the concentration level at 0.05 g/L was only included in the form of pure protein samples is therefore not representative for complex samples.

### Calibration and Preprocessing

MLR is a classical method for multivariate calibration. MLR is known to be sensitive to collinearity which is inherent in spectral data, as these do not consist of independent variables. To assert that PLS actually generates more precisely calibrated models, both approaches were tested based on identical data. The measurements were performed in 96 well MTP with 100 μL sample and PLS models were based on six latent variables. Further, it was tested whether mean centering of the spectral data has a positive effect on the calibration. For comparison, the resulting 95% confidence intervals were used (listed in Table I). The MLR model calibrated with mean centered data was shown to have a precision approximately three times lower compared to a PLS model calibrated with mean centered data. Further, mean centering of the spectral data showed to be essential in obtaining a precise model, as the confidence intervals increased by one order of magnitude when the models were calibrated based on raw data compared with the achieved precision of models calibrated based on mean centered data. Hence, the best result was achieved using mean centered data in combination with PLS.

**Table I.** Ninety-five percent confidence intervals for each protein species based on models calibrated with PLS or MLR. Each model type was based on raw or mean centered data.

| Calibration method | Applied preprocessing | Confidence interval (g/L) | | |
|---|---|---|---|---|
| | | ribA | cytC | lys |
| PLS | Mean centering | 0.031 | 0.032 | 0.028 |
| PLS | None | 0.227 | 0.349 | 0.275 |
| MLR | Mean centering | 0.097 | 0.089 | 0.102 |
| MLR | None | 0.350 | 0.590 | 0.414 |

### Method Robustness

All factors influencing the absorption in the applied range from 240 to 300 nm will have an effect on precision and accuracy of the method. Hence, it was tested whether the addition of sodium chloride has an effect on precision and accuracy of the calibration. In the tested range from 0 to 250 mM sodium chloride a significant change in precision or accuracy was not observed. Further, it was tested whether changing temperature of the sample has an influence. The tested range was 23–35°C. In a sample containing equal amounts of all three proteins, the determined lys concentration exhibited a decreasing tendency and the determined ribA and cytC concentrations exhibited a increasing tendency, both with increasing temperature. However, the change was not significant.

## Application for Process Development

The applicability of the established analytical assay for selective protein quantification was tested on an adsorber screening performed in HTE mode. The screening including four different cation exchanger resins (CM Ceramic HyperD F, Toyopearl SP 650 M, SP Sepharose FF, and Fractogel EMD $SO_3$) and generated a total of 96 samples. The objective of the screening was to determine which adsorber would yield the best separation of ribA, cytC, and lysozyme. After determining the protein concentrations in the fractions, an asymmetric Gaussian function was fitted to the data points. In Figure 5A the resulting chromatograms are displayed. Mass balances and chromatographic resolutions were calculated based on the fitted curves. For comparison, the protein concentrations in the fractions were also determined using analytical chromatography. Based on both analytical approaches, the chromatographic resolutions and the mass

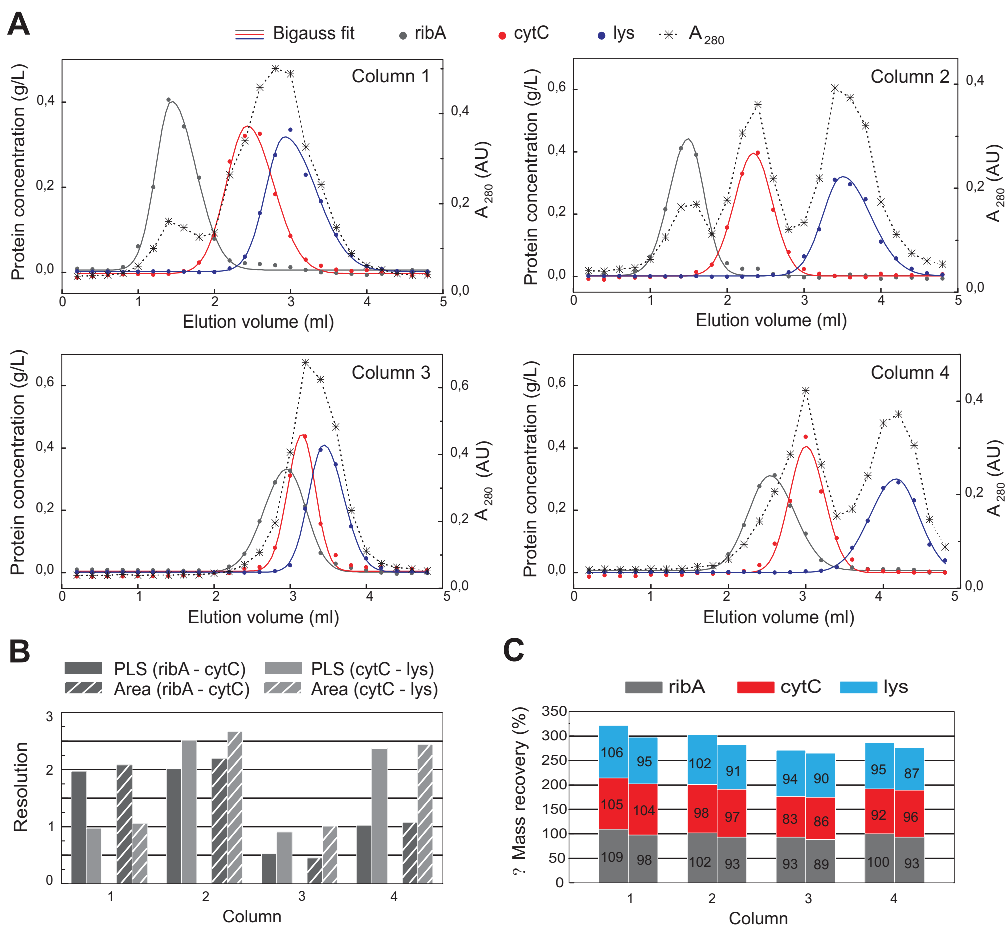


**Figure 5.** Results of the resin screening. Column 1: SP Sepharose FF, column 2: Toyopearl SP 650 M, column 3: CM Ceramic HyperD F, column 4: Fractogel EMD $SO_3$. **A**: Resulting chromatogram for each applied resin type. For each fraction, the concentration of the three protein species and the total absorption signal is displayed. The concentrations were determined using the fast absorption spectra based assay. Each data point represent the determined concentration using the fast absorption spectra based assay. An assymetrical Gauss function was fitted to the data points. **B**: Based on the fitted curves the chromatographic resolution of ribA-cytC and cytC-lys was calculated for both analytical approaches. **C**: Based on the fitted curves the mass balances for each column were calculated for both analytical approaches. [Color figure can be seen in the online version of this article, available at http://wileyonlinelibrary.com/bit]

balances were calculated. The results hereof are displayed in Figure 5B and C. A tendency of marginally higher resolutions was observed when using analytical chromatography for protein quantification. This can be explained by the higher precision of the analytical chromatographic assay at low concentrations (RSD $\sim$ 1% at 0.05–0.3 g/L). RibA and cytC were separated with similar resolution ($R \sim 2$) on column 1 and 2. CytC and lys were separated with similar resolution ($R \sim 2.5$) on column 2 and 4. Only column 2 was able to separate all three proteins and column 3 was not able to separate any of three proteins under the applied conditions. Despite the mentioned differences in the determined resolutions, both analytical approaches delivered the same overall result of resin 2 (Toyopearl SP 650 M) being the best suited and resin 3 (CM Ceramic HyperD F) being the least suited resin for the separation of the three proteins. The mass balances for each of the protein species were calculated based on both quantification methods (Fig. 5C). PLS based multivariate calibration resulted in mass balances in the range 83–109% and analytical chromatography resulted in mass balances 86–104%. This result corresponds with the higher accuracy of the chromatographic assay. Further, a tendency of a lower overall recovery was observed when using the chromatographic assay. In general, the mass balances are very acceptable for this type of screening. The advantages of the presented fast multivariate assay including the high throughput (>3 samples per min), the non-invasive and label-free methodology which is important in case the material/proteins are needed for further experiments, and the easy automation far outweighs the slightly lower precision compared to the assay based on analytical chromatography.

## Conclusion and Outlook

An assay capable of non-invasive and label-free selective protein quantification in protein mixtures was successfully established. The thorough assay validation showed that it generates very precise results. PLS was shown to generate more precise calibrations when compared to MLR and further mean centering of the spectral data was shown to be essential for precision. To demonstrate the applicability of the assay, it was used as analytics for an adsorber screening performed in HTE mode. The yielded results were highly similar to those achieved by analytical chromatography, hence it can be concluded that a high reduction in analysis time without loss of analytical information was possible. The presented methodology for selective protein quantification therefore holds potential to help eliminate the analytical bottleneck which has been created by the enhanced experimental throughput in downstream process development. In future work, the impact of the number of proteins and their spectral characteristics on the precision of the method will be investigated. Further, possibilities in the area of in-line application as process analytics for preparative chromatography will be pursued.

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